

Progress Report - 2026-08-03

From One Shared Belief to a Network of Them

Rizqi Ramadhan

2026-08-03

Item	Content
Name	Rizqi Ramadhan
Meeting Date	2026/08/03
Previous Meeting Date	2026/07/13

This document accompanies the slide deck for today's 201 seminar (`04_meetingNotes/seminarDeck/2026-08-03_progress-201-seminar-deck.md`). The deck carries the claim; this report carries the argument, the term definitions, and the reproducibility trail behind each slide.

1. What I Want to Discuss Today (Most Important)

1. **What the network results imply for survey design.** The headline result today (Section 7, Experiment 3) is that estimating with the wrong network is worse than not modelling the network at all. I want to discuss whether this means the network-generation question — name generator versus Kajiyama-san's GPS co-movement approach — should become the immediate next priority, ahead of the 72_I_595 digest (the newtown supermarket research).
2. **How to combine micro and macro interaction.** Sensei raised this on July 13 and I have not yet returned to it. I do not have a concrete proposal; I want direction on whether it is a Paper 1 question now or a later one.

2. Current Situation (Progress Since Last Meeting)

What I did

- **Read the review paper properly and located my model in its taxonomy.** Confirmed that 74_I_715.pdf is Sensei's own review, and that the "group is a special case of network" result in it resolves the apparent five-way fork from the July 13 seminar into one map. Digest at 98_attachments/from_sensei/74_I_715_digest.md.
- **Built a full teaching deck for Lee, Li & Lin (2014)** — the paper Sensei named as the next cell in the taxonomy — using the /teaching-deck skill: analogies first, the paper's own contraction-mapping bound derived by hand, a worked example, and a code walk-through. File: 98_attachments/lee_2014_network_choice/2026-07-27_lee-2014-teaching-deck.md (~117 slides, 15 sections).
- **Rebuilt the toy model as a network model in R**, one function per equation, in a new R/ folder inside the existing model_toy_two-worlds repository:
 - lee2014_core.R — the shared engine: network builders, the contraction bound, the rational-expectations solver, the log-likelihood, an NFXP maximum-likelihood estimator, marginal effects, and the data-generating process.
 - lee2014_smoking.R — reproduces the paper's own Add Health-style application at toy scale.
 - lee2014_pudo_network.R — the PUDO toy case: 30 households, drive-vs-walk, five experiments. Full account of what each one shows is in Section 7 below.
- **Cross-checked the new code against the old model.** A complete network with homogeneous households in the new vector code reproduces the four hand-verified numbers from the July model.py validate() table to 4+ decimal places (Section 7, Experiment 0) — evidence the network code is not a new, separately-trusted thing, but a strict generalization of what was already validated.
- **Pulled estimation forward from the August plan.** The July 15 plan had deferred estimation methods (the review's §5 — CCP/NFXP/MCMC) to August, with only equilibrium simulation in scope for July. Because Lee's own methodology *is* the estimation procedure, I made an explicit, deliberate scope change to include a working NFXP estimator now rather than defer it.

What I could not do / What is delayed

- **A full week was lost to an unplanned break.** After a preceding full-on working stretch I felt drained for close to two weeks and took a week off. Everything the July 15 plan (03_inbox/2026-07-15_rest-of-july-plan-v6.md) scheduled to start Monday July 20 instead started Monday July 27 — a straight one-week slip, not a change in scope.

- **This week’s Friday Horensou report did not go out.** I was attending an intensive course through Saturday and again this morning (Aug 3, 09:00–12:00), which left no Friday session. This seminar and this document are the substitute, the same call made on July 13 when the weekly report was likewise folded into the seminar document.
- **The 72_I_595 digest (Sensei’s own empirical shopping-choice paper) is still queued.** It was scheduled for the July 15 plan’s Week 1 and has not been started.
- **The estimation theory is runnable but not yet internalized.** I can execute and correctly interpret the NFXP estimator; I cannot yet re-derive the contraction-mapping bound or the likelihood construction from a blank page, the way I can for the original Brock–Durlauf model.
- **The micro/macro interaction question from July 13 is untouched.**

3. Definitions of Terms

Term	Definition
Weights matrix W (or G)	The network, written as a table: row i , column j holds w_{ij} , how much household i weighs household j ’s belief. Row-normalized so each row sums to 1; $w_{ii} = 0$.
M_i^*	Household i ’s equilibrium belief about the neighborhood, on the $[-1, +1]$ margin scale — the network analogue of the single shared m^* in my existing model.
γ (endogenous effect)	The strength of the social pull: how much a household’s belief responds to $(WM^*)_i$, its own friends’ average belief. Plays the role βJ played in the group-average model.
Contextual effect (δ)	How much a household’s belief responds to its friends’ <i>characteristics</i> (e.g. their distance to school), as opposed to friends’ beliefs.
Contraction mapping	The condition guaranteeing the belief-updating map has exactly one fixed point regardless of starting guess — Banach’s theorem, distinct from mere <i>existence</i> of a fixed point (Brouwer’s theorem).
$ \gamma < 1$	The sufficient contraction bound used throughout Section 7: below it, the rational-expectations equilibrium is provably unique.
NFXP (nested fixed-point) estimation	A maximum-likelihood estimator that re-solves the entire network equilibrium inside every likelihood evaluation, rather than approximating it. Slower but exact; the estimation method used throughout.

Term	Definition
Group vs. network structure	Group: every member responds equally to the whole group's average (a network where every tie has equal weight — a special case). Network: each member responds only to its own, generally distinct, set of ties.
Complete vs. incomplete information	Complete: an agent observes others' actual choices. Incomplete: an agent only forms a belief (a rational expectation) about others' choices, without observing them directly. My model and Lee's are both incomplete-information.

4. My Own Thoughts

On discussion item 1 (network-generation priority)

- **Current thought:** Experiment 3's result — that a misspecified network is worse than no network model at all — is, to me, an argument for prioritizing the network-generation question (survey name generator vs. GPS co-movement) over the 72_I_595 digest (the newtown supermarket research), since without a credible network the whole Lee-et-al. apparatus cannot honestly be applied to real PUDO data.
- **Reasoning:** The result is not a general methodological point I am asserting from the literature; it is something I generated myself, on my own toy PUDO network, this week — true network $\hat{\gamma} = 0.772$ against a true 0.60, versus a wrong-network $\hat{\gamma} = 1.115$ with the contextual distance effect sign-flipped ($0.283 \rightarrow -0.271$).
- **Alternatives I considered:** Digesting 72_I_595 first, since it was already queued and is Sensei's own paper. I am not confident this is the wrong call, which is why it is listed as Discussion Item 1 rather than decided unilaterally here.

5. Where I Am Stuck / What I Cannot Decide on My Own

- **Type: no clear criterion.** Whether to prioritize the network-generation survey-design question or the 72_I_595 digest next. Both are legitimate next steps; I do not have a principled way to rank them myself.
- **Type: not enough information.** How to combine micro (household-level) and macro (neighborhood-network-level) interaction, as raised on July 13. I have not identified even a candidate framework for this yet.
- **Type: too many options.** Whether the network-generation process should be a survey name generator or Kajiyama-san's GPS-based inference (I have not read his thesis yet, so this is currently a named choice without enough basis to choose).

6. Things I Want to Confirm

- Is the taxonomy reading in Section 1 (Discussion Item 1) the correct interpretation of the July 13 discussion?
- Is pulling estimation forward from August into this week's deliverable (rather than keeping it equilibrium-only, as the July plan originally scoped) a reasonable scope change?
- Should I obtain Kajiyama-san's thesis before or after digesting 72_I_595?

7. Computational Results / Figures

All results below come from `R/lee2014_pudo_network.R` in the `model_toy_two-worlds` repository (fixed random seed; ~20 seconds to run end to end; nothing here is invented or hand-adjusted). Code link: https://github.com/rizqrama/research_model_toy_two-worlds-problem (branch `main`; the `R/` folder was written locally on 2026-07-27 — **note for Sensei: not yet pushed as of this writing**, will be pushed the morning of Aug 3 before the seminar).

Result	Link to Code	Notes
Experiment 0 — cross-check vs. <code>model.py</code>	<code>R/lee2014_pudo_network.R</code> , lines ~26–49	Complete network + homogeneous households reproduces the four hand-verified <code>model.py</code> numbers to 4+ decimals: +0.005050, +1.000000, −0.999909, +1.000000.
Fig. <code>pudo_bd_vs_lee_spread.png</code> (Experiment 1)	same file, ~lines 71–93	Same 30 households, $\alpha = 0.7$: complete network mean $p^* = 0.680$ (SD 0.191); sparse school-gate network mean $p^* = 0.461$ (SD 0.264) — the network alone triples the spread.

Result	Link to Code	Notes
Fig. pudo_tipping_survival.png (Experiment 1b)	same file, ~lines 95–131	Path-dependence (two-worlds tipping) reappears on a network, first at $\beta = 1.80$ (two cliques), 2.20 (sparse school-gate), 2.50 (well-mixed random) — all well past the old $ \gamma < 1$ bound.
Fig. pudo_structure_sweep.png (Experiment 2)	same file, ~lines 134–159	Fixed households, fixed $\beta = 1.5$, four network shapes: ring (mean 0.700, SD 0.348), random (0.908, 0.147), two cliques (0.934, 0.054), hub (0.977, 0.020).
Fig. pudo_misspecified_network.png (Experiment 3)	same file, ~lines 162–210	True network recovers $\hat{\gamma} = 0.772$ (true 0.60), true-network $\hat{\delta}_{\text{distance}} = 0.283$ (true 0.20); wrong (complete) network gives $\hat{\gamma} = 1.115$ and flips the sign of $\hat{\delta}_{\text{distance}}$ to -0.271 , with an implausibly small standard error.

Action Items for Next Meeting (fill in after the meeting)

- []
- []
- []